

ACRME SKU-Level Placement WHAT-IF — User Guide (Phase 2)

Workbook: `acrme_sku_placement_whatif.xlsx`

Builder: `build_acrme_sku_scoring_whatif.py`

Baseline: Azure Capacity & Quota Management Consolidated Requirements Baseline v2.4

Status: Mockup / feasibility — derived planning figures only; not connected to live Azure state. Does **not** modify the v2.4 baseline or the Phase 1 workbook.

1. What this is (and how it differs from the region-selection what-if)

This is a **new, standalone** what-if — a different kind of analysis from `acrme_region_selection_whatif_acrme_regions.xlsx`.

	Region-selection what-if	This — SKU-level placement what-if (Phase 2)
Capacity grain	ONE blended number per region	Per SKU (CAP-022/023)
Quota grain	ONE number per environment/region	Per VM/quota family (QUA-002/003/004)
Availability term (α)	blended region free-pool	per-SKU <code>GROUP AVAIL = MIN(reserved-free, family quota-available)</code>
Question answered	“Which region overall?”	“For this SKU, which region for Prod / CVAL / DR — and is it READY?”
Readiness	region-level	RDY-002 per candidate

Phase 2 is the design-doc step “availability-by-group in scoring: point α at per-SKU group-availability; emit RDY-002 readiness per candidate.” It is **read-only** — the freeze & logical-lock layer is **Phase 3** and is not in this file.

2. Workbook structure (processing order)

Order	Sheet	Role	Editable
1	ReadMe	In-workbook overview	No
2	Request	The placement request	Yellow cells
3	Policy	Score weights & thresholds	Yellow cells
4	SKU_Family_Map	SKU → quota family lookup	No
5	SKU_Catalogue	Managed SKU set (vCPU, family, flags)	No
6	Capacity_By_SKU	Reservation plane: Region × Env × SKU	No (rebuild)
7	Quota_Groups	Quota plane: pooled group per Region × Family	No (rebuild)
8	Score_Prod	SKU-grain placement score — Production	No — formulas
9	Score_CVAL	SKU-grain placement score — CVAL / Non-Prod	No — formulas
10	Score_DR	SKU-grain placement score — Disaster Recovery	No — formulas
11	Result	Recommended Prod/ CVAL/DR + ranked candidates	No — formulas

Data flow: Request + Policy → (read against) Capacity_By_SKU + Quota_Groups → Score_Prod / Score_CVAL / Score_DR → Result . Every computed cell is a live formula; edits propagate instantly.

3. How to run a scenario

1. **Request sheet** (yellow): set **Geography** (US/EU dropdown), **Requested SKU** (dropdown of the 8 modelled SKUs), **VM count**, **Customer ID** . The sheet derives **Quota Family**, **vCPU/instance**, and **Requested cores** (= VMs × vCPU).

2. (Optional) **Policy sheet** (yellow): tune weights `w_alpha` / `w_beta` / `w_epsilon` (must sum to 1.00 — check `B7`) and thresholds `risk_factor`, `min_buffer_floor`.
3. Read **Result**: the recommended region per environment, its PS and readiness, the overall readiness, and the ranked Prod-candidate table.

One SKU per run. Placement is evaluated at SKU grain, so a multi-SKU workload is run once per SKU.

4. The scoring model (per candidate region, for the requested SKU)

For each of the 8 candidate regions, in each environment, the sheet pulls the region's numbers for the **requested SKU** from `Capacity_By_SKU` / `Quota_Groups` via `SUMIFS` keyed on (Region, Environment, SKU) and (Region, Family):

Column	Meaning	Formula (essence)
Reserved / Reserved-Free	reservation for this SKU	<code>SUMIFS(Capacity_By_SKU!J / K, ...)</code>
Family Q-Limit / Q-Avail	pooled quota for the family	<code>SUMIFS(Quota_Groups!E / G, ...)</code>
GROUP AVAIL	placeable headroom	<code>=MIN(Reserved-Free, Family Q-Avail)</code>
InGeo	region geography = requested geography	<code>=IF(B=Request!Geo,1,0)</code>
Meets-Cap	GROUP AVAIL $\geq$ requested cores	<code>=IF(GROUP AVAIL $\geq$ Req Cores, 1,0)</code>
Readiness (RDY-002)	per-candidate state	see §5
Eligible	<code>InGeo AND Meets-Cap</code>	<code>=IF(AND(InGeo,Meets-Cap), 1,0)</code>

Score components (only for eligible candidates):

- **α (availability fit)** = `1 - requested cores / GROUP AVAIL` — the **headroom cushion left after placement**. Higher = safer. This is the SKU-grain change — α is driven by the per-SKU group availability, not a blended region pool. (Chosen over a capped `MIN(avail/req,1)` because that saturates at 1 and fails to discriminate between candidates; the cushion form ranks regions by how much room remains.)
- **β (quota headroom)** = `family quota-available / family quota-limit`.
- **ϵ (buffer safety)** = `reserved-free / reserved`.

Placement Score: `PS = w_alpha· $\alpha$  + w_beta· $\beta$  + w_epsilon· $\epsilon$` (defaults 0.50 / 0.30 / 0.20).

Rank: dense rank of PS among eligible candidates (Rank 1 = highest PS). The **top pick** per environment is `INDEX/MATCH` on Rank = 1; if no region is eligible it shows `NONE ELIGIBLE`.

5. Readiness states (RDY-002, per candidate)

State	Condition
RESERVATION_DEFICIT	reserved-free $\leq$ 0, or capacity is the binding shortfall when GROUP AVAIL < requested
QUOTA_DEFICIT	family quota-available $\leq$ 0, or quota is the binding shortfall when GROUP AVAIL < requested
READY_WITH_RISK	eligible but GROUP AVAIL < requested cores $\times$ risk_factor (thin cushion)
READY	eligible with comfortable cushion

The **Binding Constraint** column shows whether CAPACITY or QUOTA is the limiter for that row.

6. Worked examples (verified with a headless recalc — 0 formula errors)

Scenario A — Standard_E32ads_v5 , 6 VMs (192 cores), US:

- **Prod** → East US 2 (GROUP AVAIL 684, READY) — the only US region with enough per-SKU headroom; Central US and Canada Central fall to RESERVATION_DEFICIT , West US 3 to QUOTA_DEFICIT (greenfield, no quota provisioned).
- **CVAL** → ranks **East US 2 (α 0.978) > Canada Central (0.877) > Central US (0.650)** — the cushion-based α discriminates correctly.
- **DR** → East US 2, READY_WITH_RISK (DR is a 30% bootstrap, so cushions are thin).

Scenario B — Standard_D8as_v5 , 2 VMs (16 cores), US:

- **Prod** → East US 2 (1191) > Central US (613) > Canada Central (301); all READY . DR now READY too (small request fits the bootstrap).

7. Data provenance & rulings baked in

- **Allocated** cores are derived from real usage (SKU_USage.xlsx). **Canada Central** is a flagged mock (0.5 $\times$ mean of the other US regions — it is absent from the source data). **DR = 30% of Prod** (ENV-005 bootstrap, not a full duplicate).
- **West US 3** shows deficits across SKUs by design: the source data has ~0 observed allocation there, so its quota limit is 0 — the correct signal for a reserved-but-unprovisioned greenfield region.
- Rulings from Phase 1: one **pooled** quota group per region+family across Prod+NonProd+DR (QUA-004); ENV-003 hard separation applies to **capacity reservations only** (quota pooled — separate planes); quota **hoarded** to one pool and distributed on demand (QUA-003).

8. Limitations / deferred

- **Read-only feasibility** — does not modify the v2.4 baseline or the Phase 1 workbook.

- **One SKU per run** (SKU grain).
 - **CRG scope is regional** (`crg-...-reg`); per-AZ CRGs (CAP-023, `crg-...-az1`) are a later increment.
 - **Freeze & logical lock** (the `LOCKED` state + `Allocation_Ledger` , logical-vs-physical CAP-002 distinction) is **Phase 3** — not in this file.
 - **SKU→family map is naming-derived**; a canonical Azure family table would replace `SKU_Family_Map` .
 - `total_customers` (γ distribution term) remains undefined in v2.4 — not used by this model, still an open baseline issue.
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Phase 2 SKU-grain placement what-if — baseline v2.4 · created 24 Sep 2026